Ogbonna Arinze

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SUMMARY

Lead Full-Stack Engineer with 8 years of professional experience building scalable, production-grade web applications across healthcare, fintech, civic tech, and enterprise B2B environments. Deep expertise in **TypeScript**, **React**, **Node.js**, and **Next.js**, with hands-on backend proficiency in **NestJS**, **RESTful API design**, **PostgreSQL**, and **Prisma ORM**. Proven technical leader who has mentored engineering teams, driven architecture decisions, established code quality standards, and delivered CI/CD-driven development workflows. Passionate about applying technology to improve healthcare outcomes, with direct experience building clinical data platforms at Genentech (Roche).

CORE COMPETENCIES

- **Full-Stack Development:** TypeScript, JavaScript (ESNext), React, Next.js, Node.js, NestJS, and Vue.js.
- **Backend & Data:** RESTful API design and implementation, Prisma ORM, PostgreSQL, MongoDB, database schema design, JWT authentication, and RBAC.
- **Frontend Architecture:** Scalable component systems, design systems, state management (Redux, Zustand, React Query), SSR/SSG, and monorepos (Nx).
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA pipelines.
- **Cloud & Infrastructure:** Docker, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Vercel, and Nx monorepo orchestration.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, Sentry, Datadog RUM, and bundle analysis.
- **Leadership:** Team leadership and mentoring, technical architecture reviews, code reviews, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product for connecting and transferring site data to upstream EHR systems, migrating from a **Vue.js** web application to a **React** Chrome extension with **TypeScript**, optimizing clinical site workflows and enhancing adoption.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend using **TypeScript** and **React Query**, with HTTP-only cookies for secure token handling and permission-based UI rendering, collaborating closely with backend engineers on auth strategies.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- **Mentored a team of two junior developers** (interns), conducting regular code reviews and fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- **Led the design and implementation** of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the **React** application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.

- Established **engineering standards** for code quality, testing, and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Utilized **Storybook** for documenting **React** components, streamlining change management and supporting agile project management practices.
- **Mentored junior engineers** and guided frontend best practices, fostering a culture of high-quality, maintainable code and consistent design across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, translating business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- **Mentored two interns** through frontend and backend projects, providing technical guidance, task management, and career development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB database schemas** and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out framework-managed chunk tables from Prisma's drift detection.
- Built **NestJS RESTful APIs** with comprehensive **database schema design** and data modeling, implementing phone OTP authentication, session management, and **role-based access control** for the admin dashboard.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring 22 interactive chart types auto-generated from AI responses, dark mode, URL state management, and responsive mobile layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and graceful shutdown.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*